Marine seismology data/metadata standards

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FDSN Working Group 5 Marine Seismology Standards Action group:

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Updates since proposed (6 June 2026)

- 24 June 2026: Changed <DriftCorrection><Type> values to match msmod inputs (e.g. “PiecewiseLinear” -> “piecewise_linear”)

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1 Introduction

This document is the result of two years of discussion about how to correctly process and transmit marine seismology data and metadata within the framework of FDSN seismology standards. We present the results here, divided into two sections:

- **Propositions:** for proposed changes to existing FDSN standards and their documentation, and
- **Recommendations and Reminders:** for recommended processing and preparing methods that will allow instrument facilities to create data and metadata that respects these standards.

A Note on clock corrections

One of the major challenges with distributing marine seismology data is the lack of a timing reference during the deployment period, which can result in time offsets of hundreds of samples by the end of the mission. In the following sections we distinguish between two types of data:

1. **"NOT CLOCK CORRECTED"**: No time correction applied. May be preferred by users of long-period data (>10s) because it can be easier to concatenate daily files.
2. **"CLOCK CORRECTED"**: Indicate the time correction in each record header and apply it. Allows the user to work with time-corrected but otherwise unmodified data.

We recommend providing "CLOCK CORRECTED**"** data.

Some of the recommendations rely on a new (in 2026) version of msmod that will allow direct clock drift and leap second corrections.

2 Proposed changes to FDSN standards and documentation

We propose that the following be added to the following reference standards or their documentation.

2.1 Source identifiers

We recommend the following additions to the documentation of FDSN Source Identifiers (https://docs.fdsn.org/projects/source-identifiers/en/v1.0/channel-codes.html#source-and-subsource-codes).

2.1.1 Subsource codes

2.1.1.1 Seismometer channels

If a component's relative angles to another component are known but neither component's absolute angles are known, the angle values should be the relative angles and `plusError` and `minusError` should be set to 180.

For example, for horizontal channels 1 and 2 oriented as recommended below:

- 1: `<Azimuth unit="DEGREES" plusError="180" minusError="180">0</Azimuth>`
- 2: `<Azimuth unit="DEGREES" plusError="180" minusError="180">90</Azimuth>`

We recommend that, if there are two orthogonal horizontal channels that are not known to be aligned to N and E, they be named 1 and 2, with 2 oriented 90° clockwise of 1 when viewed from above.

We recommend that, if the vertical channel is inverted (NOT RECOMMENDED), use subsource code 3.

2.2 StationXML metadata

The following proposals are with respect to StationXML 1.2, documented at https://docs.fdsn.org/projects/stationxml.

2.2.1 New `<ClockDrift>` element

We propose a new `<Station>`-level `<ClockDrift>` element, based on `<Equipment>`. The standard `<Equipment>` elements are used to describe the instrument's time base. Additional elements are:

`<NominalDriftRate>`
Required: no
Content_type: double
Description: Nominal drift rate (s/s) of the time base

`<Reference>`
Required: no
Content type: string
Description: Describes the time reference (for example, "GNSS", or "cross-correlation with station G:TAM")

`<DriftCorrection>`
Required: no
Note: If this element is omitted, the instrumentation is known to have significant clock drift but this drift has not been measured
Content-type: object
Sub-elements:

`<Type>`
Required: yes
Content_type: string
Description: One of:

- "piecewise_linear": time corrections are linearly interpolated between the <Sync> values
- "cubic_spline": time corrections are cubic spline interpolated between the <Sync> values
- "polynomial {a0} {a1} {a2} [{a3} [{a4} [{a5} …]]]": time corrections are calculated as: new_time = old_time + a0 +a1*deltaT + a2*deltaT**2 + a3*deltaT**3 + …, with deltaT = old_time – the first <Sync><Instrument>.

`<Sync>`
Required: yes
Many: yes
Content_type: string
Description: Instrument-to-reference synchronization. Each sub-element represents the instrument and reference datetimes recorded simultaneously.
Note: There is no extrapolation for "piecewise_linear" and "cubic_spline" types: only times between the first and last <Sync> elements are quantified.
Sub-elements :
`<Instrument>` required

Required: yes
Content type: datestring (ISO8601 yyyy-mm-ddTHH:MM:SS[.F*] format, where the number of decimal places should not be greater than those of the sampling interval, followed by "Z" to specify UTC)
Note: All times are assumed to be leap second-corrected, if necessary.

`<Reference>` required
Content type: datestring (same format as for `<Instrument>`)

Example:

```
<ClockDrift>
	<Description>MCXO Time Base</Description>
	<Manufacturer>Seascan, Inc</Manufacturer>
	<Model>SISTMB Ver4.1E</Model>
	<NominalDriftRate>1e-8</NominalDriftRate>
	<DriftCorrection>
		<Reference>GNSS</Reference>
		<Type>piecewise_linear</Type>
		<Sync>
			<Instrument>2025-01-02T03:00:00Z</Instrument>
			<Reference>2025-01-02T03:00:00Z</Reference>
		</Sync>
		<Sync>
			<Instrument>2025-11-01T02:00:00Z</Instrument>
			<Reference>2025-11-01T02:00:00.034Z</Reference>
		</Sync>
	</DriftCorrection>
</ClockDrift>
```

2.2.2 New `<Leveler>` element

The instrument's leveling system can affect the quality and absolute values of measurables.

We propose a new `<Station>` or `<Channel>`-level `<Leveler>` element, based on `<Equipment>`. The standard `<Equipment>` elements describe the leveler hardware. Additional elements are:

`<Threshold>`:
Required: yes
Content type: double
Range: 0.0 <= Threshold <= 90.0
Description: Deviation from vertical which triggers releveling
Example: `<Threshold>2.0</Threshold>`
Attributes :
`unit`:
Required: no
type: string
description: The type of unit being used. This value is fixed to be DEGREES, setting it is redundant.

`<Accuracy>`:
Required: yes
Content type: double
Range: 0.0 <= Accuracy <= 90.0
Description: Maximum deviation from vertical accepted after relevel
Example: `<Accuracy>0.1</Accuracy>`

Attributes :

unit:

Required: no
type: string
description: The type of unit being used. This value is fixed to be DEGREES, setting it is redundant.

`<MaxRelevelInterval>`

Required: no
Content type: double
Range: 0.0 <= MaxRelevelInterval
Description: Longest interval between level checks during the deployment
Example: `<MaxRelevelInterval>24</MaxRelevelInterval>`
Attributes :

unit

Required: no
type: string
description: The type of unit being used. This value is fixed to be HOURS, setting it is redundant.

`<NumRelevels>`

Required: no
type: integer
Range: 0 <= NumRelevels
Description: Number of relevels during the deployment
Example: `<NumRelevels>5</NumRelevels>`

Example:

```
<Leveler>
       <Description>Gimbal with bubble level</Description>
       <Manufacturer>Nanometrics, Inc</Manufacturer>
       <Model>T120 Compact v6.2</Model>
       <Threshold unit="DEGREES">2.0</Threshold>
       <Accuracy unit="DEGREES">0.1</Accuracy>
       <MaxRelevelInterval unit="HOURS">168</MaxRelevelInterval>
</Leveler>
```

2.3 miniSEED data

The following are recommended additions to miniSEED documentation (https://www.fdsn.org/pdf/SEEDManual_V2.4.pdf and https://fdsn-docs.readthedocs.io/projects/miniseed3/en/latest/index.html).

"**FSDH**" refers to the miniSEED 2/3 Fixed Section Data Header.

"**EH**" refers to a miniSEED Extra Header with the specified path

2.3.1 Clock drift correction

(Note that the Time Correction is ADDED to the original instrument time.)

The miniSEED 3 rules are direct translations of the miniSEED 2 rules following the recommended mapping from miniSEED 2.4 (https://fdsn-docs.readthedocs.io/projects/miniseed3/en/latest/appendix.html#miniseed-2-4-fixed-section-data-header-fsdh).

"NOT CLOCK CORRECTED" data:

- miniSEED 2:
  - Set the FSDH data quality indicator to "D".
  - Verify that FSDH activity flag (field 12) bit 1 (“time correction applied”) equals 0
- miniSEED 3:
  - Set the FSDH Data publication version to 2
  - Verify that EH.FDSN.Time.Correction equals “0” or does not exist

"CLOCK CORRECTED" data:

- miniSEED 2:
  - Set the FSDH data quality indicator to "Q".
  - Set FSDH field 16 (“Time Correction”), to the correction (in units of 0.0001 seconds).
  - Set FSDH activity flags (field 12) bit 1 (“time correction applied”) to 1.
- miniSEED 3:
  - Set the FSDH Data publication version to 3
  - Set EH.FDSN.Time.Correction to the correction (text, units of float seconds)
  -

We recommend using the msmod “--cc” command (available in v1.3.1+) to correct clock drifts. This command correctly sets the “Time Correction” record header “Time Correction” and the miniSEED 2, “time correction applied” flag..

2.3.2 Unmeasured clock drift

If clock drift is expected, but not measured/estimated:

- miniSEED 2:
  - Set the FSDH data quality indicator to "D"
  - Set the FSDH data quality flags (field 14), bit 7 ("time tag is questionable") to 1.
- miniSEED 3:
  - Set the FSDH Data publication version to 2
  - Set the FSDH Flags (field 3) bit 1 ("time tag is questionable") to 1

2.3.3 Leap second correction

Leap seconds should not be corrected in "**NOT CLOCK CORRECTED**" data.

Leap seconds MUST be corrected in "**CLOCK CORRECTED"** data.

Note: An upcoming msmod command will allow you to correct leap second(s) that occurred during your deployment.

3 Recommendations and Reminders

The following recommendations and reminders are intended to help collectors of marine seismology data and manufacturers of dataloggers to prepare their data and metadata so that they can be easily distributed and used by other researchers. Following these recommendations should allow the greatest compatibility with seismology data from land stations and from marine stations collected by other groups.

3.1 Source identifiers

The following are source identifier rules for pressure sensors (https://docs.fdsn.org/projects/source-identifiers/en/latest/channel-codes.html#pressure):

- Set Dip=90. if the recorded value DECREASES for a pressure increase.
- Set Dip=-90. if the recorded value INCREASES for a pressure increase.
- Set the subsource code to “H” if the sensor is a hydrophone
- Set the subsource code to “G” if the sensor is a differential pressure gauge
- Set the subsource code to “O” if the sensor is an absolute pressure gauge / bottom pressure recorder

3.1.1 Station names for repeated deployments

The IASPEI Station Coding Standard (https://www.isc.ac.uk/registries/download/IR_implementation.pdf, page 5) recommends that station and/or location codes be changed if the associated sensors are moved far enough to result in a significant ***teleseismic*** travel-time residual discrepancy. We recommend the same, except that the basis for what is a significant travel-time residual discrepancy should depend on your experiment.

If you change the station name between deployments, we recommend incrementing the last *N* characters of the station name, e.g. ``STAA``, ``STAB``, ``STAC``, or ``STA01``, ``STA02``, ``STA03``, etc. The value of *N* depends on the maximum number of deployments you will possibly make and the characters you use in the incrementor (numeric, alphabetic, or alphanumeric).

3.2 StationXML metadata

3.2.1 Subnetwork files

We recommend using obsinfo subnetwork files (https://obsinfo.readthedocs.io/en/latest/subnetwork.html) to store essential information about OBS deployments. These files can help in creating StationXML files containing this information.

3.2.2 Adding non-standard information

If you have information that you consider important to add to metadata, you should propose a new element for StationXML. Exceptions are information that changes frequently throughout the deployment (see if it can fit into the miniSEED file, the proposed GeoJSON format or the under-discussion Annotation format). If your information is relevant to StationXML and your proposal has not yet been adopted, the information MAY be embedded inside a `<Comment><Value>` element as JSON. The `<Comment> subject` attribute SHOULD be set to "`ProposedElement; application/json`" to indicate that the comment contains a JSON representation of a proposed future StationXML element. The JSON object SHOULD contain a single top-level key whose name is the proposed element name, and its value SHOULD follow the structure of the proposed future StationXML element as closely as possible.

Below is an example of embedding the proposed `<ClockDrift>` element:

```
<Comment subject="ProposedElement; application/json">
  <Value>{
    "ClockDrift": {
      "Description": "MCXO Time Base",
      "Manufacturer": "Seascan, Inc",
      "Model": "SISTMB Ver4.1E",
      "NominalDriftRate": 1e-8,
      "DriftCorrection": {
        "Reference": "GNSS",
        "Type": "piecewise_linear",
        "Syncs": [
          {
            "Instrument": "2019-05-12T07:47:00Z",
            "Reference": "2019-05-12T07:47:00Z"
          },
          {
            "Instrument": "2019-09-28T05:45:00Z",
            "Reference": "2019-09-28T05:45:00.034Z"
          }
        ]
      }
    }
  }</Value>
</Comment>
```

3.2.3 Input units

For pressure sensor channel responses, `InputUnits` should be `Pa`, without a `Description`

3.2.4 Deployments in lakes

We remind that the `<WaterLevel>` elements should be set to the elevation of the lake surface during the experiment.

3.2.5 Uncertainty in station/channel positions, dips and azimuths

We recommend using the `plusError`, `minusError` and `measurementMethod` attributes to specify uncertainties in `Latitude`, `Longitude` and `Elevation`, plus estimated `Dip` and `Azimuth`, and how these values were measured/estimated.

3.3 miniSEED data

3.3.1 Data completeness

We recommend keeping all of the recorded data, including "noisy" or "bad" sections.

3.3.2 Correcting drift in data that contains a leap second

An upcoming msmod version will be able to add leap seconds to data, see its help.

We suggest that the clock drift correction be performed before the leap second correction, to avoid the possibility of the leap second being shifted outside of the record containing the leap second flag.

Note: if you use the data to calculate the drift rate, you will need to first generate LEAP SECOND-CORRECTED data from the "NOT CLOCK CORRECTED" DATA. After using this data to calculate the clock drift, throw this LEAP SECOND-CORRECTED data away and process as suggested above.

3.3.3 Sample rate in miniSEED 3

Specify the actual sample rate (for example 249.9999999923 sps) in each FSDH (this is not possible in miniSEED 2, which can only use 16- or 32-bit words for the sample rate)

3.4 Other recommendations

3.4.1 Geophone wiring and polarity

The default polarity on short-period geophones manufactured for the exploration industry (e.g. the Geospace GS-11D) is such that a positive output voltage represents downward ground motion, in contradiction to the standard followed in passive seismology where a positive output voltage represents upward ground motion. We recommend that all ground-motion sensors follow the passive seismology standard. For geophones, this can be readily accomplished by connecting the positive (negative) terminal of the geophone to the negative (positive) inputs to the data-logger. This can be done on the geophone itself or by switching the leads from the geophone at the data-logger inputs.

3.4.2 Saving processing information

We recommend that processing done on data files (from data download to delivery to the data center) should be recorded in text-based, structured files. The JSON process-steps format (https://github.com/WayneCrawford/sdpchainpy?tab=readme-ov-file#process-stepsjson) is an example.